

Sara Zahedi Movahed

EDUCATION

- **Saarland University** Saarbrücken, Germany
MSc in Computer Science Oct 2025 – Present
- **Sharif University of Technology (SUT)** Tehran, Iran
BSc in Computer Engineering Sep. 2019 – Jul. 2024

RESEARCH EXPERIENCE

- **UKP Lab, TU Darmstadt** Remote
Research Assistant (HiWi) Mar 2026 – Apr 2026
 - **Research:** Conducted research on large language models (LLMs), with a focus on efficient inference and evaluation under resource constraints.
 - **LLM Inference & Optimization:** Executed and analyzed transformer-based models across configurations, improving inference efficiency through memory-aware strategies and precision tuning.
 - **Evaluation Framework Adaptation:** Extended the CheckEval framework by designing task-specific evaluation questions and aligning it with project requirements.
 - **Dataset Annotation & Manual Evaluation:** Annotated datasets and systematically evaluated model outputs by scoring multiple responses per query to assess quality and identify top-performing models.
 - **Model Analysis:** Compared open-source LLMs in terms of feasibility, performance, and resource utilization in a GPU cluster environment.
 - **Tech Stack::** Python, PyTorch, Hugging Face Transformers, vLLM, Slurm, GPU Computing, Large Language Models
- **Imperial College London** Remote
Research Assistant Sep 2023 – Sep 2025
 - * **Research:** Worked as a Research Assistant on a conversational psychotherapy chatbot project at the intersection of machine learning, human-computer interaction, and mental health research.
 - * **Conversational AI System:** Contributed to the development of a psychotherapy chatbot that used machine learning techniques to identify user problems and recommend relevant therapeutic exercises based on Self-Attachment Techniques.
 - * **Frontend Development:** Designed and implemented the chatbot user interface using React.js, with a focus on conversational flow and usability, and collaborated with backend services developed using Django.
 - * **Experimental Evaluation:** Participated in designing and running user studies to experimentally evaluate the effectiveness and user experience of the chatbot system.
 - * **Research Output:** The project resulted in a peer-reviewed archived publication, with an additional paper currently in preparation for submission to a CHI poster.
 - * **Tech Stack::** React.js, Django, Machine Learning, Conversational AI, Human-Computer Interaction
- **Deakin University & RMIT University** Remote
Research Assistant Sep 2023 – Dec 2025
 - * **Research Collaboration:** Contributed to a multi-institutional research project investigating accessibility issues in low-code applications, focusing on challenges faced by users with visual impairments.
 - * **Dataset Construction:** Constructed and curated a large-scale dataset of 4,762 low-code application reviews by collecting, preprocessing, and manually annotating data from multiple commercial review platforms.
 - * **Hybrid ML Model:** Designed and implemented a hybrid classification model combining Transformer-based sentence embeddings with a keyword-based system to detect accessibility-related issues in user reviews.
 - * **Model Evaluation:** Evaluated the proposed model against multiple baseline Transformer models (BERT, RoBERTa, DistilBERT), achieving the best performance with 78.07% accuracy and 78.52% F1-score on the test set.
 - * **Performance Analysis:** Demonstrated that the hybrid approach significantly improves recall (82.70%) over standalone Transformer models by capturing accessibility-related patterns missed by deep learning models alone.
 - * **Research Output:** This work was published at the International Conference on Evaluation and Assessment in Software Engineering (EASE 2025), ACM.

- * **Tech Stack::** Python, Transformers, NLP, Deep Learning, PyTorch, Data Annotation, Accessibility Research

Sharif University of Technology

Tehran, Iran

Research Assistant

Oct 2023 – Mar 2024

- * **Democratic Content Moderation (CHI LBW 2024):** Conducted research on decentralized and democratic social media policy-making, analyzing disparities between users' content moderation preferences and platform-level policies.
- * **User Study & Data Collection:** Designed and ran large-scale user studies with 110 participants from diverse backgrounds, collecting over 3,000 interaction records across 546 tweets in multiple content categories.
- * **Web-Based Experiment Platform:** Developed a data-collection website mimicking real-world social media interfaces using React.js and Django to gather user votes, preferences, and interaction metadata.
- * **AI-Based Policy Model:** Proposed and implemented a Transformer-based policy learning model with context-aware multi-task learning and uncertainty estimation, achieving 79.82% accuracy in predicting users' preferred moderation policies.
- * **Analysis of LLMs & Bias:** Evaluated and compared content moderation policies suggested by multiple large language models, demonstrating their misalignment with democratic user preferences and limitations as decision-makers.
- * **Research Output:** This work was published as a Late-Breaking Work paper at the ACM CHI Conference on Human Factors in Computing Systems (CHI 2024).
- * **Tech Stack::** Python, Transformers, Deep Learning, NLP, React.js, Django, User Studies, Human-Centered AI

The Institute of Science and Technology Austria (ISTA)

Klosterneuberg, Vienna

Research

Jul 2022 – Sep 2022

- * **Research Assistant:** Worked as a Research Assistant in the areas of physics simulation and computer graphics, contributing to research-focused software and visualization workflows.
- * **Rendering & Simulation:** Supported physics simulation and computer graphics pipelines with a focus on efficient rendering and visualization of complex simulation data.
- * **Mesh Management:** Maintained and optimized large-scale rendering meshes using the Magnum graphics engine, ensuring performance and stability when handling a high volume of meshes.
- * **Technical Focus:** Worked on improving rendering performance and resource management for graphics-intensive research environments.
- * **Tech Stack::** C++, Magnum, Computer Graphics, Physics Simulation

EXPERIENCE

cafebazaar.ir

Tehran, Iran

Software Engineer (Frontend)

Jul 2025 – December 2025

- * **User-Facing Applications:** Developed and maintained scalable user-facing features using React, Nuxt.js, and Vue 3 across multiple projects, focusing on performance, usability, and maintainability.
- * **Frontend Architecture:** Implemented responsive and accessible UI components, managed application state, and integrated frontend logic with backend services to deliver production-ready features.
- * **Testing & Quality:** Wrote and maintained unit tests to ensure code reliability, reduce regressions, and support long-term maintainability of frontend applications.
- * **Security Considerations:** Applied frontend security best practices during implementation, including safe data handling, validation, and secure integration with backend APIs.
- * **Tech Stack::** React, Nuxt.js, Vue 3, TypeScript, JavaScript, Unit Testing, Git

international.cafebazaar.ir

Tehran, Iran

Software Engineer

Jul 2024 – Jul 2025

- * **International CaféBazaar Platform:** Worked as a Software Engineer on the international.cafebazaar.ir platform, contributing to the design and implementation of core user-facing features and system functionality.
- * **Frontend Development:** Implemented the landing page and user interface using React.js, focusing on responsiveness, usability, and clean component structure.
- * **Authentication System:** Developed login and registration flows, including verification code handling and confirmation email delivery as part of a secure authentication process.
- * **Email Automation:** Implemented an automated email system for account verification and user notifications, integrated with backend services.
- * **APK Management:** Built functionality to receive APK uploads and display submitted applications within user profiles.
- * **Infrastructure & Tools:** Worked with Django for backend integration, managed version control using Git, and deployed services using Nginx with database integration.
- * **Tech Stack::** React.js, Django, Git, Nginx, Database Systems

SKILLS

- **Machine Learning & NLP:** Transformer-based Models (BERT, RoBERTa, DistilBERT), Text Classification, Hybrid ML Models, Feature Engineering, Model Evaluation (Precision, Recall, F1)
- **Research & Experimentation:** Experimental Design, User Studies, Data Annotation, Error Analysis, Baseline Comparison, Ablation Studies
- **Scientific Communication:** Academic Writing, Peer-reviewed Publications, Paper Submission and Review Process
- **Tools & Frameworks:** Python, PyTorch, React.js, Django, Git